AMARNATH S

MECHANICAL ENGINEER

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PROFILE SUMMARY

Mechanical Engineering undergraduate with hands-on experience in CAD design, assembly modelling, and basic simulation projects.Gained industry exposure through internships in maintenance engineering and automotive engine manufacturing, and demonstrated independent design ability through CAD-based engine and mechanism projects. Seeking to build a long-term career in Mechanical Design and R&D.

EDUCATION

B.E. Mechanical Engineering — Government College of Engineering, Salem *Present | CGPA: 8.32*

Higher Secondary Education — St. Joseph's Matriculation Higher Secondary School, Hosur *2023 | 93%*

INTERNSHIP

Schaeffler Group, Hosur

- Observed preventive and breakdown maintenance activities under senior engineers
- Gained exposure to motors, bearings, gearboxes, and hydraulic systems
- Learned TPM, 5S practices, maintenance procedures, and safety standards

Ashok Leyland, Hosur

- Completed a mechanical engineering internship at Ashok Leyland, Hosur, observing diesel engine assembly across multiple platforms (H Series, P15, ZD30)
- Gained exposure to engine assembly workflows, quality control checkpoints, and production processes in a high-volume automotive manufacturing environment
- Studied design and engineering features across light, medium, and heavy-duty diesel engine platforms, including turbocharging and common rail fuel injection systems

PROJECTS

Tri-Star Stair Climbing Trolley | Academic Team Project

- Designed the rack-and-pinion gear system that drives the tri-star wheel cluster mechanism, enabling controlled climbing motion over stair steps
- Engineered the gear design to support an adjustable height mechanism, allowing the trolley to adapt to varying stair dimensions
- Assisted in fabricating the demo model using 3D printing technology, helping translate the gear design into a functional physical prototype
- Collaborated in a team setting to integrate gear design with the tri-star wheel assembly, contributing to a working stair-climbing trolley prototype

V6 Engine & Jet Engine Turbine Design | Personal Projects

- Designed and assembled individual components for a V6 engine and a jet engine turbine in PTC Creo, applying parametric 3D modeling and assembly constraints
- Simulated piston-crankshaft motion in the V6 model to validate kinematic behavior and verify component fit

Line Tracking Robot | Team Project

- Developed a line tracking robot using IR sensors and Arduino Uno with a CAD-designed body

CERTIFICATIONS

- PTC Creo: Certified User
- Attended the “Emerging Trends in 3D Printing Technology” workshop and presented a paper on “Regenerative Braking”, organized by Kongu Engineering College

ACHIEVEMENTS & VOLUNTEERING

- Participated in the Hackathon on Product Conceptualization and Prototyping, organized by the Naan Mudhalvan Program, and won a medal
- Kurukshetra 2025, Anna University, Guindy — Participated in the Ultimate Line Tracer Robot event
- Member, Bureau of Indian Standards (Standards Club)
- Serving as a member of the Society of Mechanical Engineering (SME) in the department
- Active participant in the National Service Scheme (NSS)

SKILLS

PTC Creo, Fusion 360, CATIA V5

ANSYS (Basic Stress and Structural Analysis)

Microsoft Office